

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions**



	AKI	ORGANIZED	ORADPRINTER	DOIVA	NCARML	CAIEN	OLANRIZEN	DRCPNITEFVS
1	CACAHIZE	DMGRNBIDB	DOONININE	\$1,900	NDEN	80		\$390
1		BS:56191MN	RBSN	39,008			303	1/90
2		BA:34191RA	D2ON	30,308			390	200
3	CBRI/KBS	BS:5418AKS	DOON	22,353			540	200
3	NAGY	BS:34191R8	GGON	22,106			669	290
4	MIMCEEIAN	BI:33/51R9	SGON	25,955			119	490
5	MANY	BI:567P100	GGON	\$5,585			349	390
10	BRANINOPA	FI:3019210	COON	99,003			498	490
19	BUA	BI:36.1921M	S3ON	89,353			83	400
17	MRKN	BI:39/9/218	XOON	39,222			39	390
18	MART	BI:50191MN	RSOIN	25,344			99	1090
19	SLAMENRD	BI:39091/19	GGON	25,395			388	560
15	MAVRTE	FI:5515850	SDON	29,980			359	760
16	N	BI:26150B9	SRON	22,183			358	390
17	NANDS	BI:30/30R0	BOON	22,396			358	90
17	ORINETUS	BI:3315R1B	GBIAN	34,360			193	1,380
18	MATIBE	BI:3619050	BSNR	58,353			\$59	\$1,20
19	MAMEOAS	BI:3418VR1	BEMR	\$E/395			839	1,600
18	MASMEERS	BI:3618V1M	BBMN	\$9,892			816	3500
17	MEBN	RI:34191170	BEMR	40,990			819	3330
14	NADX	RI:3118080	SBOIN	29,395			819	6000
13	FRSMUDIAN	RI:2415090	BEDIO	29,390			813	3090
12	MODIY	RI:28/9090	B51PR	20,395			830	2030
18	VUNM	RI:39/17890	SCON	45,290				\$990
18	M4HRS	RI:3815109	BBOK	21,3880				1679
19	RMIMEYA	BI:3055110	GRON	22,1500				1,690
19	MANFIVI	BI:19853090	SBOA	\$8,890				5630
17	MAREP	52:5699091	G2DA	\$9,902				1,788
18	NOTA			\$3,890				1683
11								1510

Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

Data Features



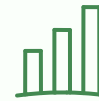
Demographics

Age, Gender, Location, Subscription Status



Purchase Details

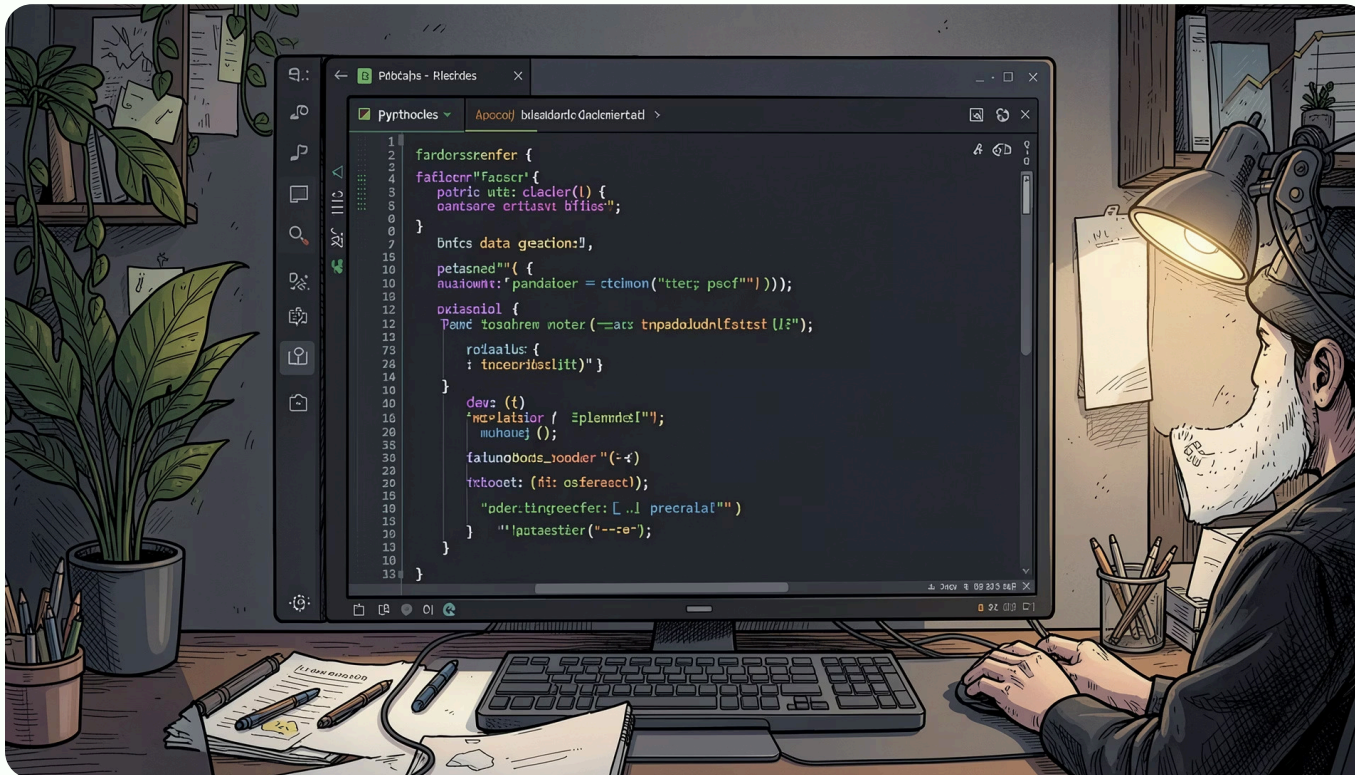
Item, Category, Amount, Season, Size, Color



Shopping Behavior

Discount, Promo Code, Frequency, Review Rating, Shipping

Python Data Preparation



→ Load & Explore

Imported via `pandas`, inspected with `df.info()` & `.describe()`

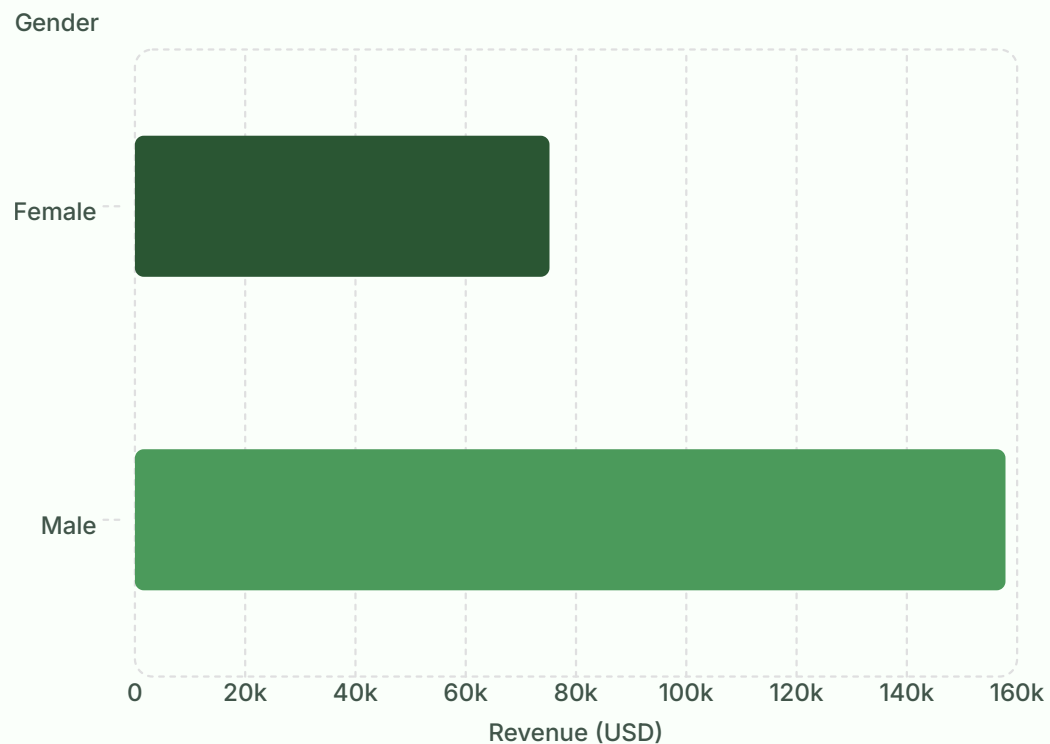
→ Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to snake_case

→ Engineer & Load

Created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`; loaded into PostgreSQL

Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers across all transactions.

📋 Male: \$157,890 · Female: \$75,191

Subscribers vs. Non-Subscribers

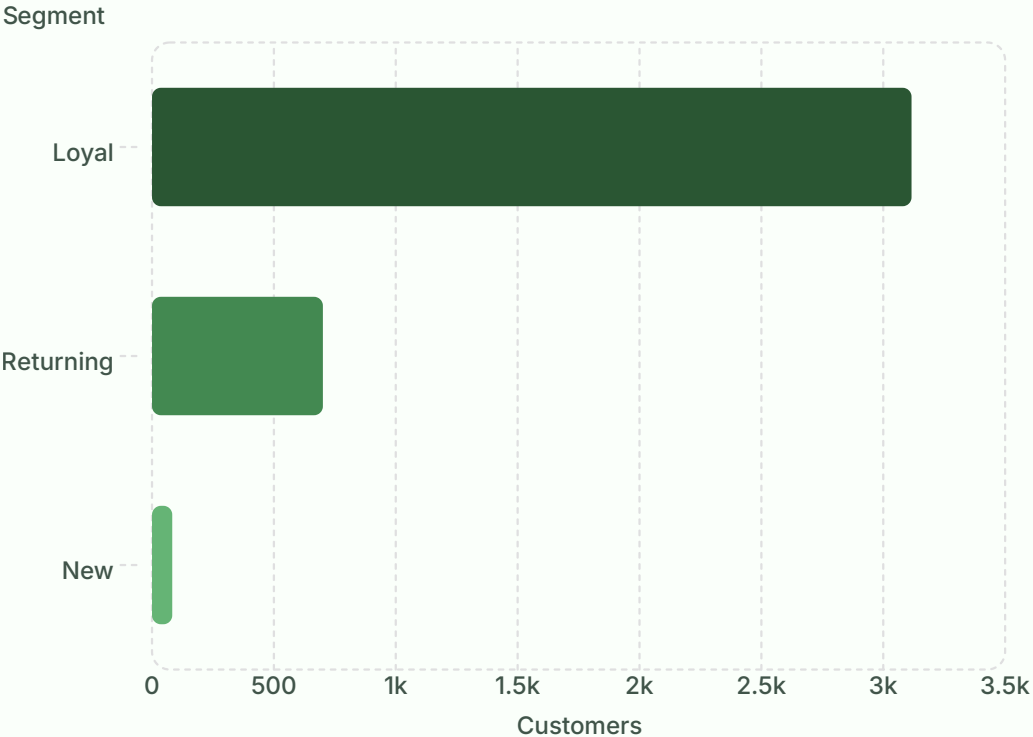
■ Customers ■ Revenue (USD)



Insight

Non-subscribers outnumber subscribers **3:1** and drive **73% of total revenue**, despite near-identical average spend (~\$59).

Customer Segmentation & Top Products



Top-Rated Products

Gloves

3.86 ★

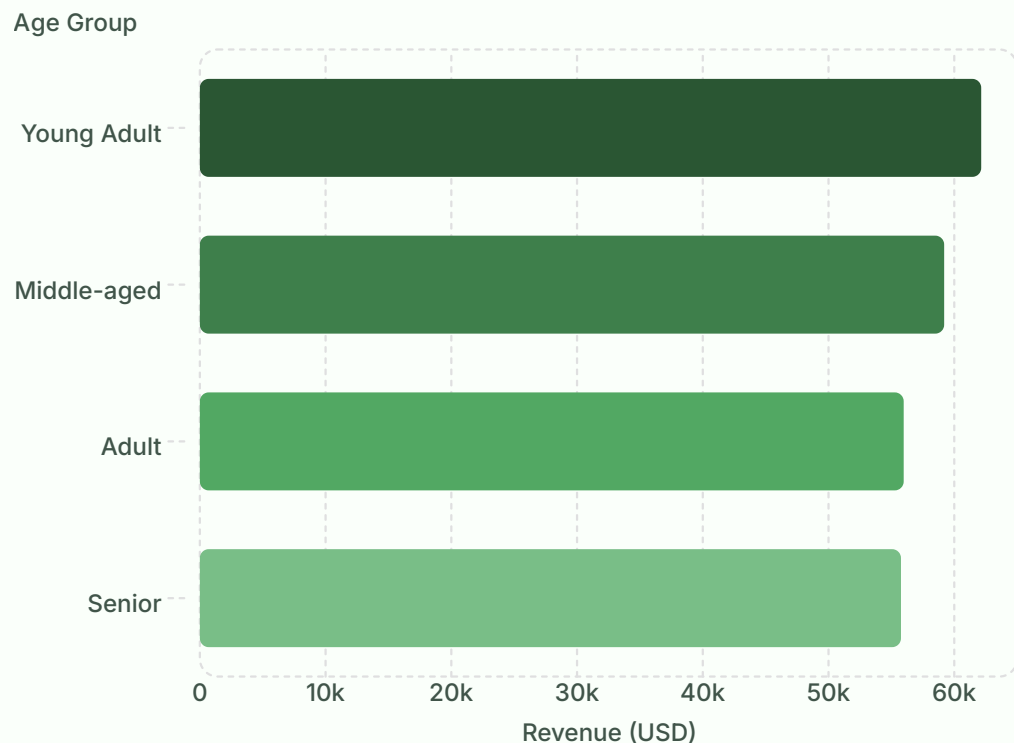
Sandals

3.84 ★

Boots

3.82 ★

Revenue by Age Group



Observation

Revenue is **surprisingly evenly distributed** across age groups – Young Adults lead by a narrow margin.

 Avg. purchase: \$50.76 · Avg. rating: 3.75



Power BI Dashboard

3.9K Customers

\$59.76 Avg Purchase

3.75 Avg Rating

Filters: Subscription Status · Gender · Category · Shipping Type

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers